Lab1_Docker

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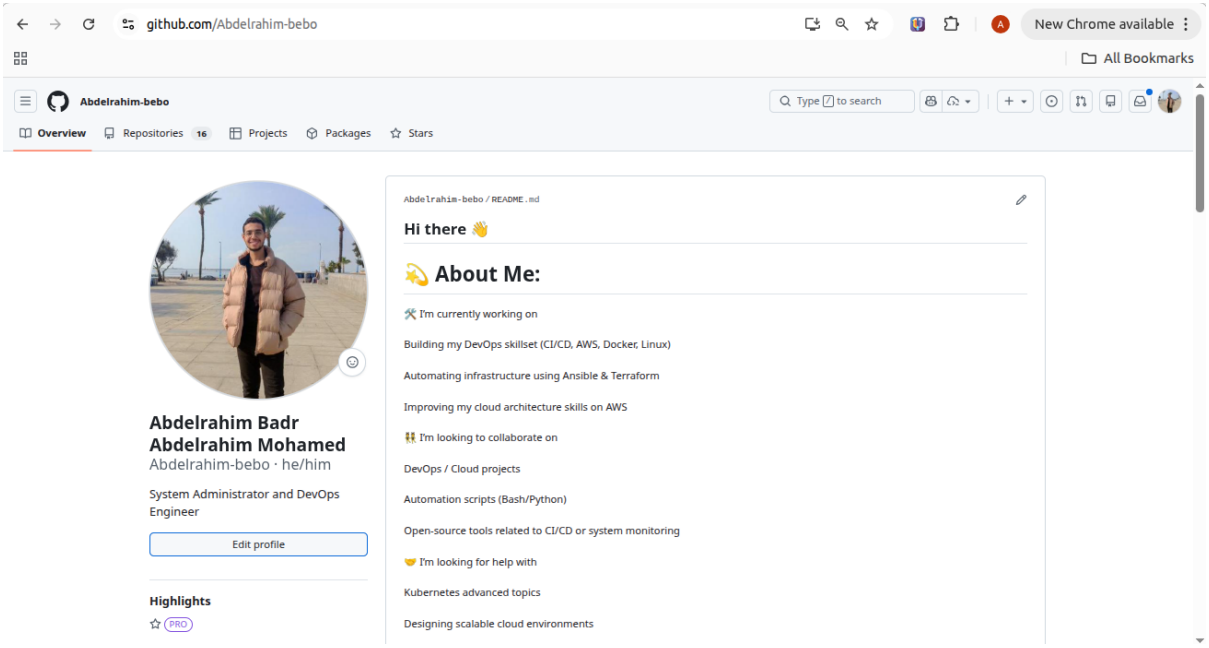
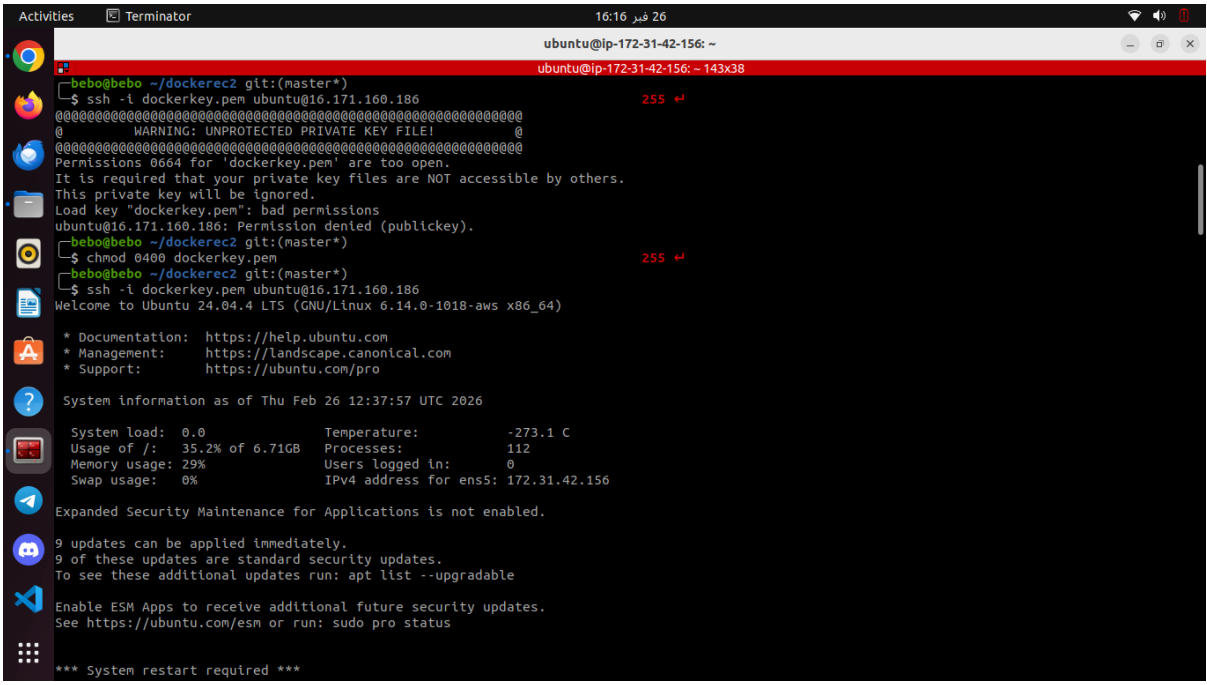
Q1:

1. The github account:



2. Downloading and installing docker on my machine:

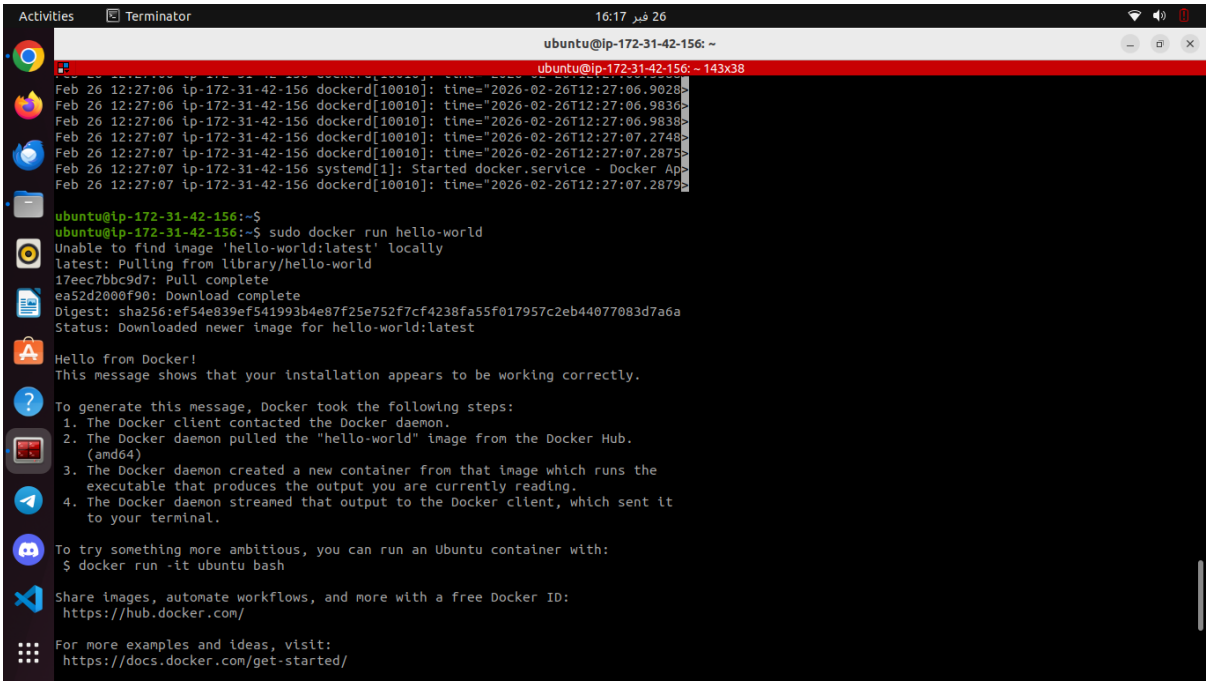
   a. First of all, i used ec2 instance to make the labs of docker on it and these are the screens of the instance


Activities
Terminator
16:16 فبر 26
ubuntu@ip-172-31-42-156: ~
ubuntu@ip-172-31-42-156: ~ 143x38
bebo@bebo ~/dockerec2 git:(master*)
$ ssh -i dockerkey.pem ubuntu@16.171.160.186
255
@@@

b. Now the downloading and installing of the docker

```
ubuntu@ip-172-31-42-156:~$
ubuntu@ip-172-31-42-156:~$ sudo apt remove $(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-doc podman-docker containe
rd runc | cut -f1)
dpkg: no packages found matching docker-compose
dpkg: no packages found matching docker-compose-v2
dpkg: no packages found matching docker-doc
dpkg: no packages found matching podman-docker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@ip-172-31-42-156:~$ # Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
```

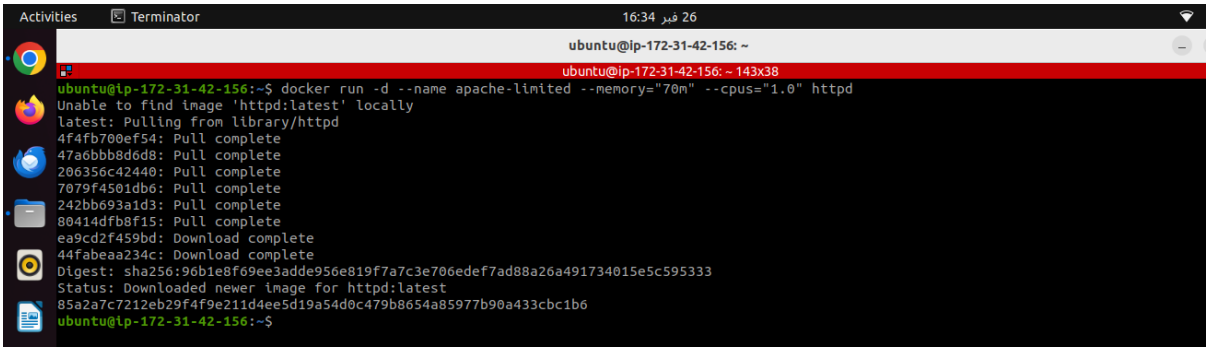
```
ubuntu@ip-172-31-42-156:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-ce is already the newest version (5:29.2.1-1~ubuntu.24.04~noble).
docker-ce-cli is already the newest version (5:29.2.1-1~ubuntu.24.04~noble).
containerd.io is already the newest version (2.2.1-1~ubuntu.24.04~noble).
docker-buildx-plugin is already the newest version (0.31.1-1~ubuntu.24.04~noble).
docker-compose-plugin is already the newest version (5.1.0-1~ubuntu.24.04~noble).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@ip-172-31-42-156:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: e>
     Active: active (running) since Thu 2026-02-26 12:27:07 UTC; 1h 34min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 10010 (dockerd)
      Tasks: 9
     Memory: 104.0M (peak: 104.9M)
        CPU: 1.091s
     CGroup: /system.slice/docker.service
             └─10010 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/con>

Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.4818>
Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.5518>
Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.5588>
Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.9028>
Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.9836>
Feb 26 12:27:06 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:06.9838>
Feb 26 12:27:07 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:07.2748>
Feb 26 12:27:07 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:07.2875>
Feb 26 12:27:07 ip-172-31-42-156 systemd[1]: Started docker.service - Docker Ap>
Feb 26 12:27:07 ip-172-31-42-156 dockerd[10010]: time="2026-02-26T12:27:07.2879>

ubuntu@ip-172-31-42-156:~$
ubuntu@ip-172-31-42-156:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
```



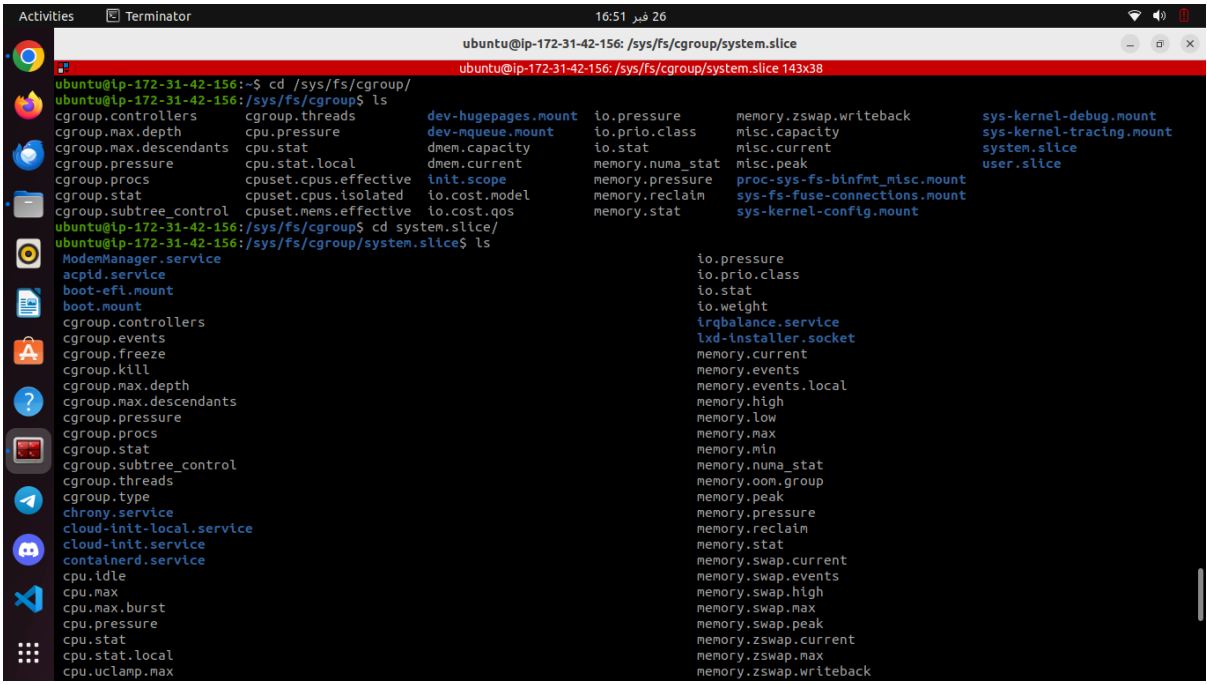
3. Pulling nginx and making it work in background with the name format MyName- ITI-<My number in ITI Sheet>

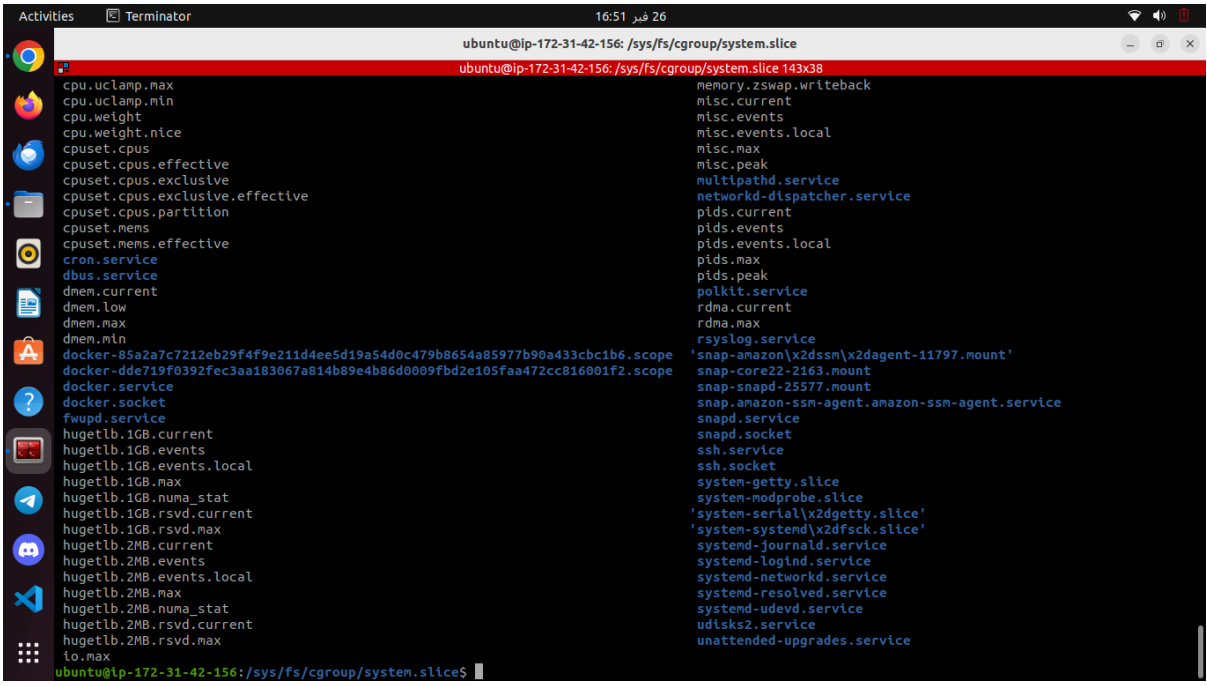
```
ubuntu@ip-172-31-42-156:~$
ubuntu@ip-172-31-42-156:~$ docker run -d --name AbdelrahimBadr-ITI-1 nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
5e7756927bef: Pull complete
589002ba0eae: Pull complete
399d0898a94e: Pull complete
955a8478f9ac: Pull complete
bca5d04786e1: Pull complete
6d397a54a185: Pull complete
3e2c181db1b0: Pull complete
6b7b6c7061b7: Pull complete
dda176b4ea00: Download complete
8a29e1f4877b: Download complete
Digest: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Status: Downloaded newer image for nginx:alpine
dde719f0392fec3aa183067a814b89e4b86d0009fbd2e105faa472cc816001f2
ubuntu@ip-172-31-42-156:~$
```

4. Hardening limits for Apache while running it

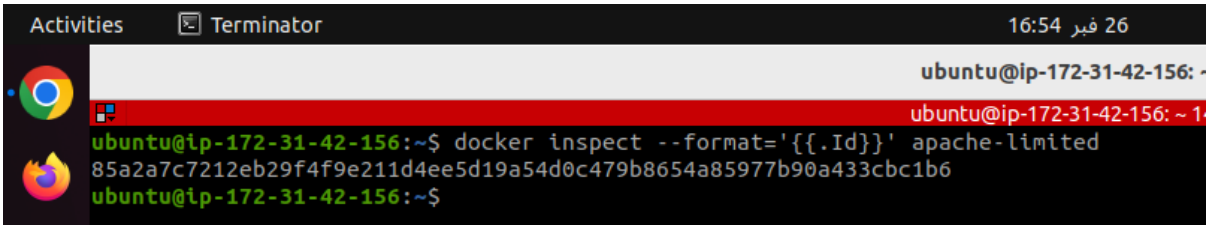


5. Make sure that docker did our rules
    a. Getting in /sys/fs/cgroup/system.slice(this is the folder of the cgroup where we find the docker-id)

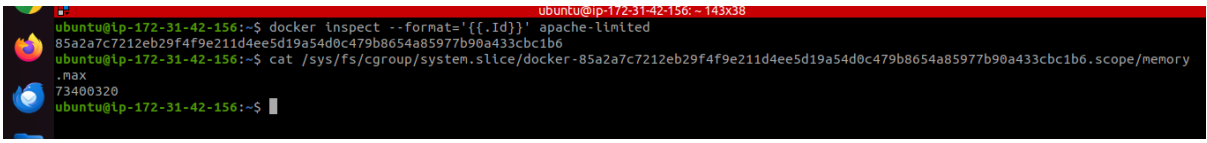




b. We found two docker ids to make sure what is exactly the one we are using now we did this command



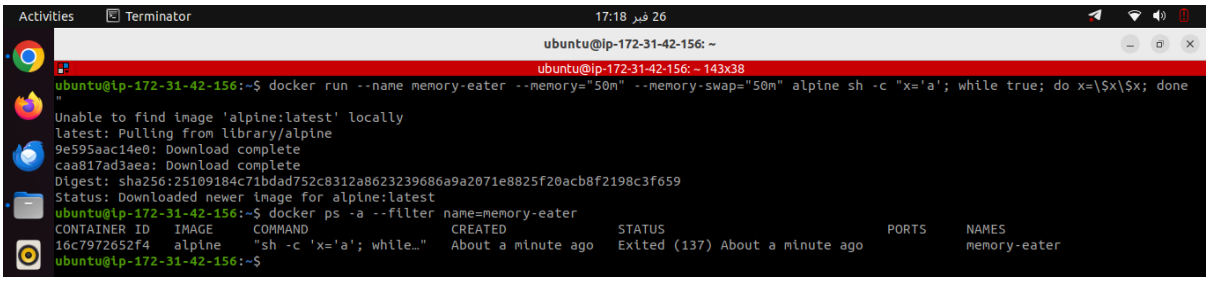
c. Seeing the maximum memory that docker used

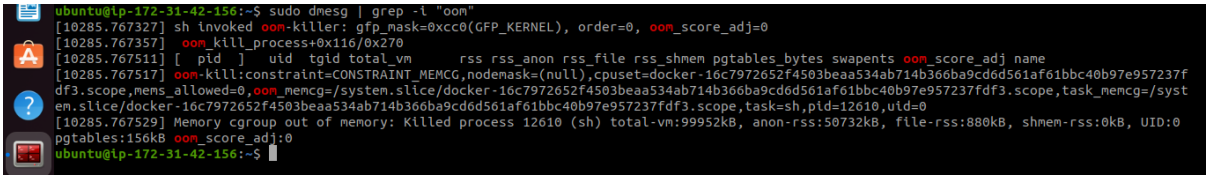


It appeared in bytes so that it’s that number.

Q2:

- What we want to do in this question is to force the memory to be increased so it doesn't follow our rules and see that this container will be terminated.
- We will make the swap equal the memory so we cancel the swap
- The infinite loop code is taking 'a' and increasing it automatically to 'aa' and 'aaa' and so on
- That is the code and what happened



What we see here is the container after being killed and when we wanted to see what happened we found Exited (137) which means SIGKILL because the memory exceeded the maximum

- The final step to prove that it's killed by oom killer we did this command